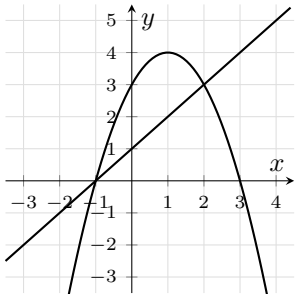
Lesson 1: Solving Linear–Quadratic Systems

Assignment

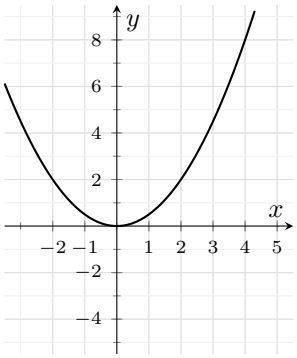
1. The graphs of $y = -x^2 + 2x + 3$ and $y = x + 1$ are shown.



(a) State the solution of the system.

(b) Verify each solution algebraically.

2. The parabola $y = \frac{1}{2}x^2$ is graphed. Graph each line on the grid, then state the number of solutions of the system and the solutions.



(a) $y = 2x$

(b) $y = 2x - 2$

(c) $y = 2x - 4$

(d) Check part (b) algebraically.

3. Solve each system algebraically. Verify one solution of each.

(a) $y = x^2 - 3x - 2$
$y = 2x + 4$

(b) $y = -x^2 + 3x + 10$
$y = x + 7$

(c) $y = x^2 + 6x + 4$
$y = 2x$

(d) $y = 2x^2 - 5x$
$x + y = 4$ (exact answers)

(e) $3x - y = 5$
$y = 3x^2 - 4x - 3$

4. Solve each system by elimination.

(a) $y = x^2 - 2x + 5$
$y = 4x - 3$

(b) $x^2 - y = 3$
$2x + y = 5$

5. Consider the system $y = x^2 + x - 4$ and $y = -x + 1$.

(a) Use a graphing calculator to solve the system, to the nearest tenth. State the window you used.

(b) Solve the system algebraically to get the exact solutions.

6. The system $y = x^2 + 3x + 6$ and $y = x + 2$ has no solution.

(a) Describe how a graph shows that a linear–quadratic system has no solution.

(b) Describe how algebra shows that a linear–quadratic system has no solution.

(c) Show algebraically that this system has no solution.

7. Without solving, determine the number of solutions of each system.

(a) $y = 3x^2 - x + 2$
$y = 5x - 1$

(b) $y = -2x^2 + x + 1$
$y = -x + 4$

(c) $y = 0.5x^2 - 2$
$y = x + 1$

8. Consider the line $y = 4x + k$ and the parabola $y = x^2 + 2x + 5$.

(a) Determine the value of k for which the line is tangent to the parabola, and the point of tangency.

(b) For which values of k does the system have two solutions? no solution?

9. The line $y = mx + 3$ is tangent to the parabola $y = 2x^2 + 5$. Determine the possible values of m and the point of tangency for each.

10. Determine the exact points of intersection of each circle and line.
(a) $x^2 + y^2 = 50$ and $y = x$ (b) $x^2 + y^2 = 17$ and $y = x + 3$

11. Show algebraically that the line $x + 3y = 10$ is tangent to the circle $x^2 + y^2 = 10$, and determine the point of contact.

12. A cross-section of a satellite dish follows $y = 0.25x^2$ and a straight support rod follows $y = x + 3$, both in decimetres. Determine the points where the rod meets the dish.

13. The solution of the system $y = x^2 - 4x + 6$ and $y = 2x - 3$ is
A. $(3, 3)$ only **B.** $(-3, -9)$ only **C.** $(3, 3)$ and $(-3, -9)$ **D.** no solution

14. The line $y = -x + k$ does **not** intersect the parabola $y = x^2 + 3x$ when
A. $k > -4$ **B.** $k < -4$ **C.** $k < 4$ **D.** $k = -4$

15. How many solutions does the system $2x - y = 3$ and $y = x^2 - 2$ have?
A. 0 **B.** 1 **C.** 2 **D.** infinitely many

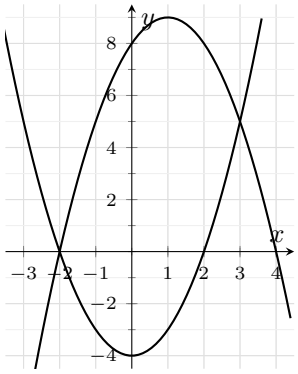
16. Numerical Response The line $y = 3x + k$ is tangent to the parabola $y = x^2 - x + 7$. The value of k is _____.

17. Numerical Response The solutions of the system $y = x^2 + 2x - 1$ and $y = 5x + 3$ are (a, b) and (c, d). The value of $a + b + c + d$ is _____.

Lesson 2: Solving Quadratic–Quadratic Systems

Assignment

1. The graphs of $y = x^2 - 4$ and $y = -x^2 + 2x + 8$ are shown.



(a) State the solutions of the system.

(b) Solve the system algebraically to confirm, and verify one solution.

2. Sketch an example of each, on your own axes, and write a pair of equations for (d).

(a) Two parabolas that both open up and have no point in common.

(b) Two parabolas, one opening up and one opening down, with exactly one point in common.

(c) Two parabolas that both open down and have two points in common.

(d) A quadratic–quadratic system with infinitely many solutions.

3. Solve each system algebraically and verify one solution.

(a) $y = x^2 + 2$
$y = 2x^2 - 2x - 1$

(b) $y = -x^2 + 5x + 1$
$y = x^2 - x - 7$

(c) $y = 3x^2 + x - 2$
$y = x^2 + 4x$

(d) $y = (x - 3)^2$
$y = x^2 + 3$

4. Solve the system $y - x^2 = 2x + 1$ and $y + x^2 = 4x + 5$.

5. Consider the system $y = x^2 - 2x - 2$ and $y = -x^2 + 2x + 3$.

(a) Use a graphing calculator to solve the system, to the nearest tenth. State your window.

(b) Solve the system algebraically to get the exact solutions.

6. Solve each system, or explain why it has no solution or infinitely many solutions.

(a) $y = x^2 - 5x + 2$
$y = x^2 + 3x - 6$

(b) $y = -2x^2 + x + 3$
$y = -2x^2 + x - 1$

(c) $y = (x + 2)^2 - 1$
$y = x^2 + 4x + 3$

7. Without solving, determine the number of solutions of each system.

(a) $y = 2x^2 + 3x + 1,\ \ y = x^2 + x$

(b) $y = x^2 - 4x + 6,\ \ y = -x^2 + 2$

(c) $y = 3x^2 - 2,\ \ y = x^2 + 5x$

(d) $y = 4x^2 + x,\ \ y = 4x^2 - 3x + 2$

8. Consider the system $y = 2x^2 + k$ and $y = x^2 - 6x$.

(a) Determine the value of k for which the system has exactly one solution, and find that solution.

(b) For which values of k does the system have two solutions? no solution?

9. Solve each system.

(a) $y = \frac{1}{3}x^2 + x - 2$
$y = -\frac{2}{3}x^2 + 2x + 4$

(b) $y = 0.5x^2 - 3x + 10$
$y = -0.25x^2 + 1.5x + 4$

10. A designer draws a logo with two arcs: $y = -0.5x^2 + 4x$ and $y = -x^2 + 7x - 4$, in centimetres. Determine the points where the arcs cross.

11. The system $y = x^2 + bx + 5$ and $y = -x^2 + 3x + 1$ has a solution with $x = 1$. Determine b and the other solution.

12. The number of solutions of the system $y = 3x^2 - 6x + 5$ and $y = x^2 + 2x - 3$ is

A. 0 **B.** 1 **C.** 2 **D.** infinitely many

13. Which system has infinitely many solutions?

A. $y = x^2 - 6x + 9$ and $y = (x + 3)^2$
B. $y = (x - 3)^2$ and $y = x^2 - 6x + 9$
C. $y = x^2 + 9$ and $y = (x - 3)^2$
D. $y = -(x - 3)^2$ and $y = x^2 - 6x + 9$

14. One solution of the system $y = 2x^2 - 3x$ and $y = x^2 + x + 5$ is $(5, 35)$. The other solution is

A. $(-1, 5)$ **B.** $(1, -1)$ **C.** $(-5, 65)$ **D.** $(-1, -5)$

15. Numerical Response The solutions of the system $y = x^2 - 2x + 7$ and $y = -x^2 + 6x + 1$ are (a, b) and (c, d). The value of $b + d$ is _____.

16. Numerical Response The parabolas $y = x^2 + 4x + k$ and $y = -x^2 + 8x + 1$ touch at exactly one point. The value of k is _____.

Lesson 3: Problems Involving Systems

Assignment

1. The sum of two numbers is 11. The square of the first number minus the second number is 19. Write a system and determine both possible pairs of numbers.

2. The difference of two positive numbers is 3 and their product is 70. Determine the numbers.

3. A rectangle has a perimeter of 46 cm and a diagonal of 17 cm. Write a system and determine the dimensions of the rectangle.

4. A rectangular swimming area at a beach is roped off on three sides, with the shoreline as the fourth side. There is 30 m of rope, and the swimming area must be 90 m^2. Let x be the width measured out from the shore.

(a) Write a linear–quadratic system that could be graphed to solve the problem.

(b) Determine the possible dimensions, to the nearest hundredth of a metre.

5. A water balloon is launched from the bottom of a slope. Its path is $y = -0.04x^2 + 1.6x$ and the slope is the line $y = 0.6x$ (metres, with the launcher at the origin).

(a) Where does the balloon land on the slope?

(b) How far up the slope is that, to the nearest tenth of a metre?

(c) What is the greatest vertical height of the balloon above the slope?

6. A parabolic arch over a walkway has the equation $y = -0.2x^2 + 8$, with the ground along the x-axis (metres). A straight support beam lies along the line $y = 0.5x + 5$ and is attached to the arch at both ends.

(a) Use a graphing calculator (*intersect*) or the quadratic formula to determine where the beam meets the arch, to the nearest hundredth.

(b) Determine the length of the beam, to the nearest tenth of a metre.

7. A small business sells x custom skateboards per month. Its revenue is $R = -0.5x^2 + 40x$ and its cost is $C = 12x + 150$, both in dollars.

(a) Determine the break-even quantities.

(b) Determine the maximum profit and the number of skateboards that gives it.

8. A drone hovers and dips over a field while a ball is kicked straight up beside it. Their heights in metres after t seconds are

$$\text{ball: } h = -5t^2 + 25t + 2 \qquad \text{drone: } h = t^2 - 5t + 38$$

(a) At what times are the ball and the drone at the same height, and what is that height?

(b) During what time interval is the ball higher than the drone? Explain.

9. Determine the value of k for which the line $y = 2x + k$ is tangent to the parabola $y = x^2 - 4x + 3$, and state the point of tangency.

10. For what values of c does the system $y = x^2 - 2x + c$ and $y = 4x - 1$ have no solution?

11. Show algebraically that the line $y = 2x - 10$ is tangent to the circle $x^2 + y^2 = 20$, and determine the point of contact.

12. An arrow is shot upward from the top of a cliff. Its height above the beach after t seconds is $h = -5(t-p)^2 + q$ metres. After 1 s it is 35 m high and after 5 s it is 15 m high.

(a) Write and solve a system to determine p and q.

(b) How high is the cliff, and when does the arrow land on the beach (to the nearest hundredth of a second)?

13. The system $y = x^2 - 6x + 10$ and $y = 2x - 6$ has

A. no solution **B.** one solution **C.** two solutions **D.** infinitely many solutions

14. The length of a rectangular deck is 4 m more than its width, and its area is 96 m^2. A system that models the deck is

A. $l = w + 4,\ lw = 96$ **B.** $l + w = 4,\ lw = 96$ **C.** $w = l + 4,\ l + w = 96$ **D.** $l = w + 4,\ 2l + 2w = 96$

15. A revenue is $R = -x^2 + 20x$ and a cost is $C = 4x + 28$. The break-even values of x are

A. 2 and 14 **B.** 4 and 7 **C.** -2 and -14 **D.** 1 and 28

16. Numerical Response The line $y = 6x + k$ is tangent to the parabola $y = x^2 + 2x + 9$. The value of k is _____.

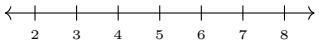
17. Numerical Response Two fireworks are launched at the same moment. One rises from the ground with height $h = -5t^2 + 35t$, the other from a 30 m roof with height $h = -5t^2 + 25t + 30$ (metres, t in seconds). The height, in metres, at which they are level with each other is _____.

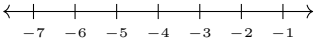
Lesson 4: Linear Inequalities in Two Variables

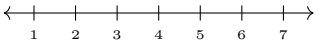
Assignment

1. Solve each inequality, check with a test value, and write the solution in set notation.

(a) $5x - 8 > 2x + 7$ (b) $4 - 3(2 - x) \leq 5x + 6$ (c) $\dfrac{2x+1}{3} - 1 < \dfrac{x}{2}$





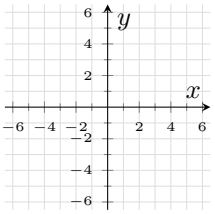


2. Determine whether each point is in the solution region of $3x + 2y \geq 6$.

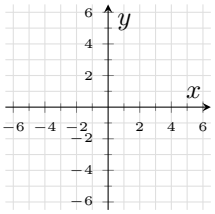
(a) $(2, 0)$ _____ (b) $(0, 0)$ _____ (c) $(1, 2)$ _____ (d) $(-1, 4)$ _____ (e) $(4, -3)$ _____

3. Graph each inequality.

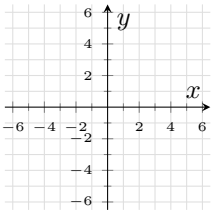
(a) $y \geq x - 4$



(b) $y < -3x + 2$

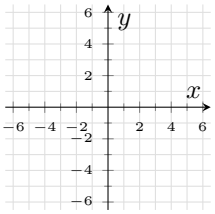


(c) $y \leq \frac{3}{4}x + 1$

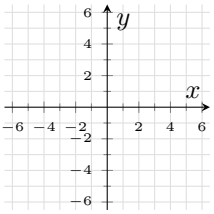


4. Graph each inequality by hand.

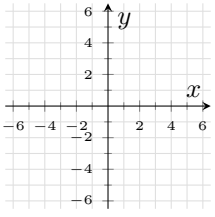
(a) $2x + 3y \leq 6$



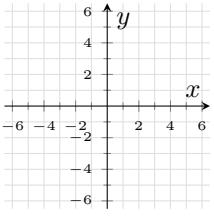
(b) $3x - 4y > 12$



(c) $3x + y < 0$

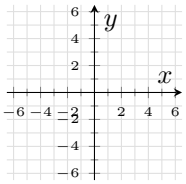


(d) $4x - 5y + 20 \geq 0$

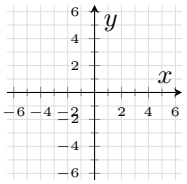


5. Graph each inequality.

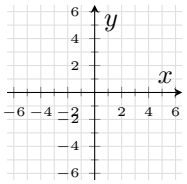
(a) $x \leq 4$



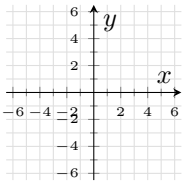
(b) $y - 1 > 0$



(c) $2x + 6 \geq 0$



(d) $-2 < x \leq 3$



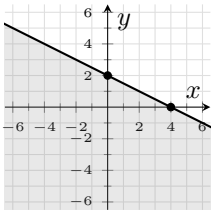
6. Maya graphed $6x - 2y \geq 8$ by rewriting it as $y \geq 3x - 4$ and shading above a solid line.

(a) Explain her error and write the correct inequality in terms of y.

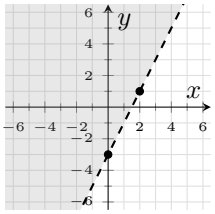
(b) Use a test point to confirm which side should be shaded.

7. Write the inequality whose solution region is shown.

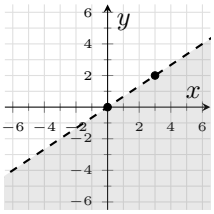
(a)



(b)



(c)

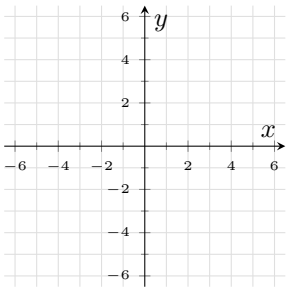


8. Determine the inequality described.

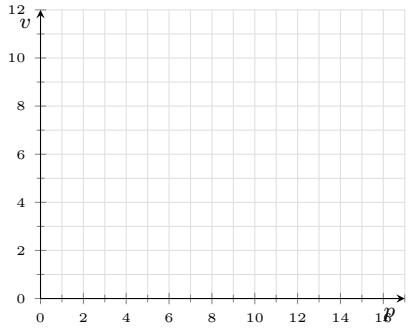
(a) The boundary line is solid with x-intercept -4 and y-intercept 6, and the solution region contains the origin.

(b) The boundary line is dashed and passes through $(-1, 5)$ and $(3, -3)$, and the solution region does not contain the origin.

9. Graph the region that satisfies both $x + y \leq 4$ and $y > x - 2$. State the point where the boundary lines meet, and whether it is in the region.



10. A club has at most \$240 for a party. Pizzas cost \$16 each and veggie trays \$24 each. Let p be the number of pizzas and v the number of trays.



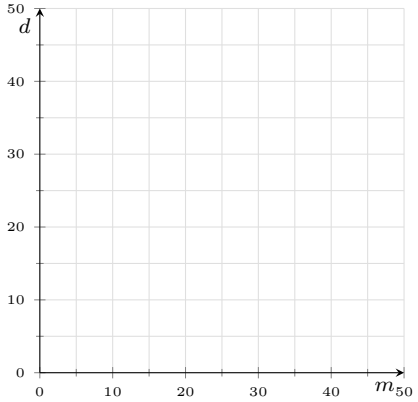
(a) Write an inequality, in simplest form, with restrictions.

(b) Graph the solution region.

(c) Can the club buy 9 pizzas and 4 trays? Explain.

(d) With 6 pizzas, what is the greatest number of trays?

11. A fitness class has room for at most 40 people. Members pay \$8 and drop-ins pay \$12, and the instructor needs at least \$360 per class. Let m be the number of members and d the number of drop-ins.



(a) Write a system of inequalities, with restrictions.

(b) Graph the solution region and label its corners.

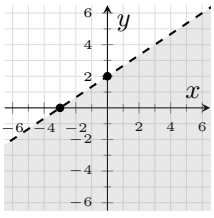
(c) Is a class with 20 members and 15 drop-ins acceptable?

(d) What is the greatest number of members the class can have?

12. The point that is **not** in the solution region of $3x - 2y \leq 6$ is

A. $(0, 0)$ **B.** $(2, 0)$ **C.** $(-2, 5)$ **D.** $(3, 1)$

13. The graph shows the solution region of



A. $2x - 3y > -6$

B. $2x - 3y < -6$

C. $3x - 2y > -6$

D. $2x - 3y \geq -6$

14. Which inequality has a solid boundary line and a solution region **above** its boundary line?

A. $y > 2x + 1$ **B.** $-2x + y \leq 1$ **C.** $4x - 2y \leq -2$ **D.** $2x - y \geq -1$

15. Numerical Response The boundary line of the solution region of $4x + by < 24$ has a y-intercept of 3. The value of b is _____.

16. Numerical Response Kai has at most \$20 to spend on pizza slices at \$4 each and drinks at \$3 each, and he buys at least one of each. The greatest total number of items he can buy is _____.

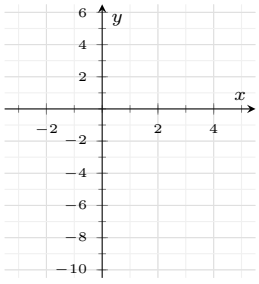
Lesson 5: Quadratic Inequalities in Two Variables

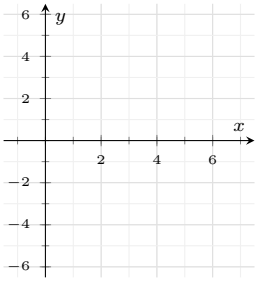
Assignment

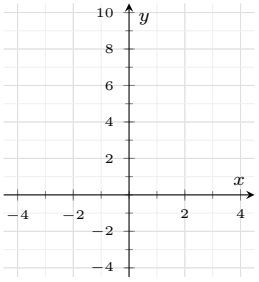
1. Determine whether each point is a solution of $y < 2x^2 - 5x + 1$. Show the substitution.
(a) $(0, 0)$ (b) $(2, -1)$ (c) $(3, 5)$ (d) $(-1, 7)$

2. For each inequality, write it with y alone on the left (if needed), state whether the boundary is solid or dashed, and whether the solution region is above or below the parabola.
(a) $y > x^2 - 7$ (b) $y \leq -3x^2 + x$ (c) $y + 4 \geq (x - 1)^2$ (d) $-y > x^2 + 2$

3. Graph the solution region of each inequality. Show the intercepts, the vertex and your test point.
(a) $y \geq x^2 - 2x - 8$ (b) $y < -x^2 + 6x - 5$ (c) $y \leq 9 - x^2$

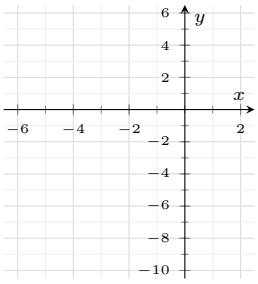




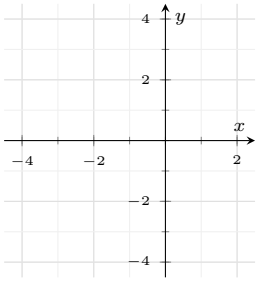


4. Graph the solution region of each inequality.

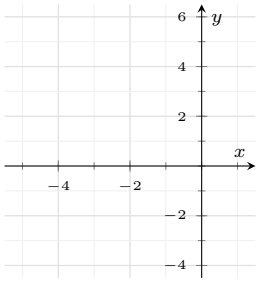
(a) $y \leq (x+2)^2 - 9$



(b) $y \geq -\frac{1}{2}(x+3)(x-1)$

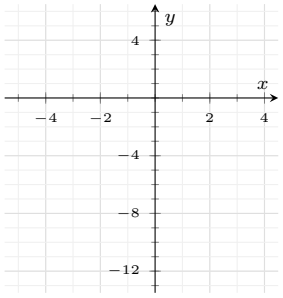


(c) $y > x^2 + 4x$

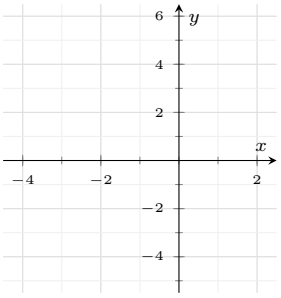


5. Write each inequality with y alone on the left, then graph its solution region.

(a) $y + 12 \leq x^2 + x$



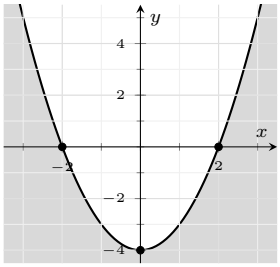
(b) $2x^2 + 4x - 2y \geq 6$



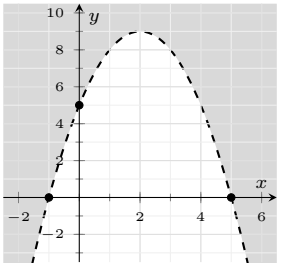
6. Explain why $(0, 0)$ cannot be used as the test point for $y < 3x^2 + 5x$. Then use another test point to decide which region is shaded.

7. Write the inequality whose solution region is shown, in general form.

(a) Vertex $(0, -4)$; x-intercepts ± 2.



(b) x-intercepts -1 and 5; y-intercept 5.



8. A solution region is bounded by the parabola with vertex $(-3, -2)$ that passes through $(-1, 6)$. The region is above the parabola and does not include the parabola.

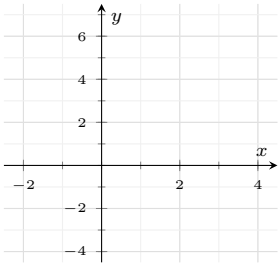
(a) Write the equation of the parabola in general form.

(b) Write the inequality.

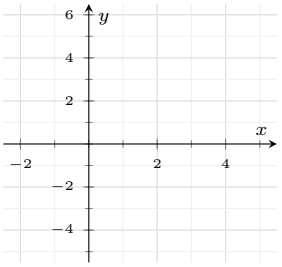
(c) Is $(-3, 0)$ a solution? Is $(0, 16)$? Explain.

9. Use a graphing calculator to graph each inequality. Sketch the region and label the intercepts (to the nearest tenth).

(a) $y > -2x^2 + 5x + 3$



(b) $y \leq x^2 - 3x - 2$



10. A lawn sprinkler sends a stream of water along the path $y = -0.4x^2 + 2x$, where x is the horizontal distance from the sprinkler and y is the height, both in metres. Points under the stream satisfy $y < -0.4x^2 + 2x$.

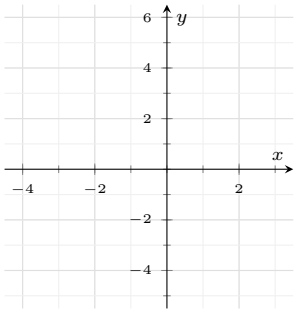
(a) How far from the sprinkler does the water land?

(b) A garden gnome 0.8 m tall stands 4.5 m from the sprinkler. Does the water pass over it?

(c) Does the water pass over a 1.5 m tall shrub 1 m from the sprinkler?

11. The solution region of a pair of inequalities is the set of points that satisfy both.

(a) Shade the region that satisfies both $y \leq -x^2+5$ and $y \geq x-1$.



(b) Find the points where the boundaries meet algebraically.

(c) Is $(0, 4)$ in the region? Is $(1, -2)$?

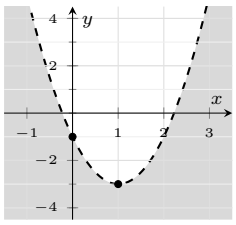
12. The boundary of the shaded region is a parabola with vertex $(1, -3)$ and y-intercept -1. The inequality is

A. $y < 2x^2 - 4x - 1$

B. $y > 2x^2 - 4x - 1$

C. $y \leq 2x^2 - 4x - 1$

D. $y < x^2 - 2x - 1$



13. Which point is in the solution region of $y \geq -x^2 + 3x + 4$?

A. $(1, 5)$ **B.** $(-1, -1)$ **C.** $(4, 0)$ **D.** $(2, 5)$

14. Written with y alone on the left, the inequality $6 - 3y > 3x^2$ is

A. $y > -x^2 + 2$ **B.** $y < -x^2 + 2$ **C.** $y < x^2 - 2$ **D.** $y > x^2 - 2$

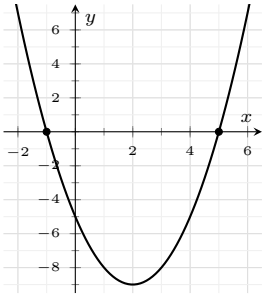
15. Numerical Response The point $(3, k)$ is in the solution region of $y > x^2 - 5x + 10$, and k is a whole number. The smallest possible value of k is _____.

16. Numerical Response The solution region of $y \leq ax^2 + bx + c$ is below a solid parabola with vertex $(2, 7)$ that passes through $(0, 3)$. The value of $a + b + c$ is _____.

Lesson 6: Quadratic Inequalities in One Variable: Roots and Test Points

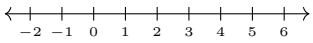
Assignment

1. The graph of $y = x^2 - 4x - 5$ is shown. Write the solution of each in set notation, and show (b) and (c) on a number line.

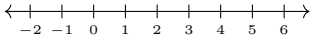


(a) $x^2 - 4x - 5 = 0$

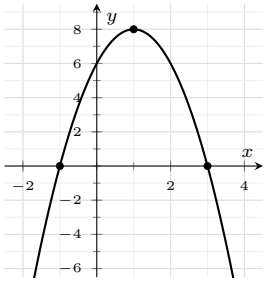
(b) $x^2 - 4x - 5 < 0$



(c) $x^2 - 4x - 5 \geq 0$



2. The graph of $y = -2x^2 + 4x + 6$ is shown. Use it to solve each inequality.

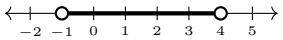


(a) $-2x^2 + 4x + 6 > 0$

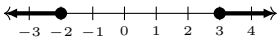
(b) $-2x^2 + 4x + 6 \leq 0$

(c) $-2x^2 + 4x + 6 \geq 6$

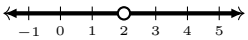
3. Write the solution shown on each number line in set notation.



(a)



(b)



(c)

4. Solve using roots and test points. Write each solution in set notation.

(a) $x^2 - 7x + 12 < 0$

(b) $x^2 + 2x - 15 \geq 0$

(c) $x^2 - 9x > 0$

(d) $4x^2 - 25 \leq 0$

(e) $2x^2 + x - 6 > 0$

(f) $-x^2 + 3x + 10 \geq 0$

5. Rearrange, then solve.

(a) $x^2 < 5x + 14$

(b) $3x^2 + 4 \geq 8x$

(c) $(x + 3)(x - 1) \leq 5$

(d) $6x^2 > 7x + 3$

6. Solve with a graphing calculator, to the nearest tenth.

(a) $x^2 - 6x + 2 < 0$

(b) $-x^2 + 4x + 1 \geq 0$

(c) $2x^2 + 3x > 7$

(d) Solve the inequality in (a) exactly, using the quadratic formula.

7. Solve each inequality.

(a) $x^2 + 6x + 9 \leq 0$

(b) $x^2 + 6x + 9 > 0$

(c) $x^2 - 3x + 5 > 0$

(d) $-x^2 + 4x - 7 \geq 0$

(e) $4x^2 - 12x + 9 < 0$

8. Find and correct the error in each student's work.

(a) To solve $x^2 \leq 9$, Sam took the square root of both sides and wrote $x \leq 3$.

(b) To solve $-x^2 + x + 6 > 0$, Priya multiplied by -1 to get $x^2 - x - 6 > 0$, and wrote $x < -2$ or $x > 3$.

(c) To solve $x^2 - x - 6 > 0$, Jordan wrote $(x - 3)(x + 2) > 0$, so $x - 3 > 0$ and $x + 2 > 0$, so $x > 3$.

9. Consider the equation $3x^2 + kx + 12 = 0$. Determine the values of k for which the equation has

(a) two distinct real roots

(b) no real roots

(c) two equal real roots

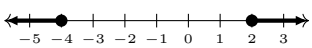
10. For what values of m does the graph of $y = x^2 + mx + m + 3$ have no x-intercepts?

11. Write and solve a quadratic inequality.

(a) The length of a rectangular garden is 3 m more than its width. For what widths is its area at least 40 m^2?

(b) The product of two consecutive integers is less than 30. Determine all possible values of the smaller integer.

12. The solution shown is the solution of



A. $(x + 4)(x - 2) \leq 0$ **B.** $(x + 4)(x - 2) \geq 0$ **C.** $(x - 4)(x + 2) \geq 0$ **D.** $(x + 4)(x - 2) > 0$

13. The solution of $x^2 > 25$ is

A. $x > 5$ **B.** $-5 < x < 5$ **C.** $x < -5$ or $x > 5$ **D.** $x \leq -5$ or $x \geq 5$

14. Which inequality has no solution?

A. $x^2 + 4 > 0$ **B.** $x^2 - 4 < 0$ **C.** $x^2 + 4 \leq 0$ **D.** $(x-4)^2 \geq 0$

15. Numerical Response The number of integers that satisfy $x^2 - 3x - 28 \leq 0$ is _____.

16. Numerical Response The solution of $x^2 + bx - 18 < 0$ is $\{x \mid -9 < x < 2, x \in \mathbb{R}\}$. The value of b is _____.

Lesson 7: Quadratic Inequalities in One Variable: Sign Analysis

Assignment

1. Solve by case analysis. Write each solution in set notation and show it on a number line.
(a) $(x+4)(x-2) < 0$ (b) $x^2 - 9x + 18 \geq 0$

2. (a) Complete the sign chart and solve $x^2 + 2x - 24 > 0$.

x					
Product					

Solve with a sign chart.
(b) $x^2 - 49 \leq 0$ (c) $3x^2 - 15x < 0$ (d) $2x^2 + 7x - 15 \geq 0$

3. Rearrange so one side is 0, then solve.
(a) $10x^2 < 3 - 13x$ (b) $4x^2 \geq 12x$

(c) $x(x-3) > 28$ (d) $(x-1)^2 \geq 2x + 13$

4. Solve. Each has a negative leading coefficient.
(a) $24 - 2x - x^2 \geq 0$ (b) $-2x^2 + 5x + 3 < 0$ (c) $-4x^2 - 4x + 3 \leq 0$

5. Solve.
(a) $x^2 + 8x + 16 \leq 0$ (b) $9x^2 - 6x + 1 > 0$ (c) $x^2 + x + 3 > 0$

(d) $2x^2 - 3x + 5 \leq 0$ (e) $x^2 + 6x + 4 < 0$ (exact, then to the nearest hundredth)

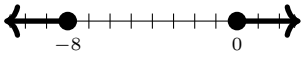
6. Solve $x^2 - 4x - 21 \leq 0$ by each method.
(a) case analysis (b) test points (Lesson 6)

(c) a sign chart (d) a graph of $y = x^2 - 4x - 21$

7. Write a quadratic inequality with integer coefficients and a positive leading coefficient that has the given solution.

(a) $\{x \mid -2 \leq x \leq 5, x \in \mathbb{R}\}$

(b) $\{x \mid x < -3 \text{ or } x > 4, x \in \mathbb{R}\}$

(c) 

(d) $\{x \mid -\frac{3}{4} < x < 2, x \in \mathbb{R}\}$

8. A stone is thrown upward from the edge of a 30 m cliff above a lake. Its height above the water after t seconds is $h = -5t^2 + 25t + 30$ metres.

(a) When does the stone hit the water?

(b) When is the stone more than 50 m above the water? For how long?

(c) When is the stone below the top of the cliff (but not yet in the water)?

(d) Show algebraically that the stone never reaches 62 m.

9. A food truck's daily profit, in dollars, when it charges x dollars per meal is $P = -5x^2 + 150x - 1000$.

(a) For what prices does the truck make a profit?

(b) For what prices is the daily profit at least \$120?

10. A rectangle has a perimeter of 40 cm. For what widths is its area at least 96 cm^2?

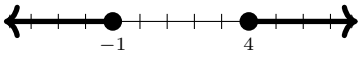
11. For what values of k does $x^2 + kx + (2k - 3) = 0$ have two distinct real roots?

12. (a) Jordan solved $x^2 - x - 6 > 0$ like this: $(x-3)(x+2) > 0$, so $x - 3 > 0$ and $x + 2 > 0$, so $x > 3$. Explain his error and give the correct solution.

(b) Maya wrote that the solution of $x^2 < 9$ is "$x < \pm 3$". Explain why this is wrong and solve the inequality.

13. *(Extension.)* Solve with a sign chart.
(a) $(x+4)(x-1)(x-3) \leq 0$ (b) $x^2(x-3) < 0$

14. The solution shown on the number line belongs to the inequality



A. $x^2 - 3x - 4 \leq 0$ **B.** $x^2 + 3x - 4 \geq 0$ **C.** $x^2 - 3x - 4 \geq 0$ **D.** $x^2 + 3x - 4 \leq 0$

15. Kai will solve $2x^2 - 7x - 4 < 0$ by testing one value of x in each interval. Which set of test values works?
A. $-1, 0, 5$ **B.** $0, 2, 3$ **C.** $-2, -1, 5$ **D.** $-1, 4, 5$

16. The solution of $12 + x - x^2 < 0$ is
A. $-3 < x < 4$ **B.** $x < -3$ or $x > 4$ **C.** $-4 < x < 3$ **D.** $x < -4$ or $x > 3$

17. Numerical Response The number of integers that satisfy $x^2 - 3x - 18 \leq 0$ is _____.

18. Numerical Response A ball is kicked upward. Its height after t seconds is $h = 49t - 4.9t^2$ metres. To the nearest hundredth of a second, the ball is more than 98 m high for _____ s.

Lesson 8: Unit 8 Review

Assignment

1. Solve each system algebraically and verify.

(a) $y = x^2 + 2x - 3,\ \ y = x + 3$

(b) $y = -x^2 + 4x,\ \ y = 2x - 8$

(c) $y = x^2 - 2x + 3,\ \ 2x - y = 1$

(d) $y = x^2 + 3,\ \ y = 2x - 1$

2. Consider the system $y = x^2 - 3x - 1$ and $y = x + 2$.

(a) Use a graphing calculator to solve the system, to the nearest hundredth.

(b) Determine the exact solution algebraically.

3. Solve each system.

(a) $y = x^2 - 4x + 6$
$y = -x^2 + 2x + 2$

(b) $y = 3x^2 + 2x - 5$
$y = 3x^2 - 4x + 7$

(c) $y = 2x^2 - 8x + 3$
$y = x^2 - 2x - 6$

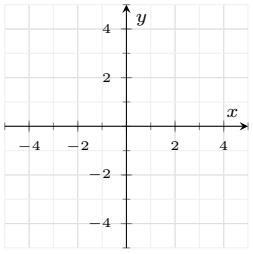
4. The line $y = kx + 2$ is tangent to the parabola $y = x^2 - 2x + 6$. Determine the possible values of k and the point of tangency for each.

5. The larger of two numbers is 3 more than the smaller. The square of the smaller number plus the larger number is 23. Write a system of equations and solve it to determine the numbers.

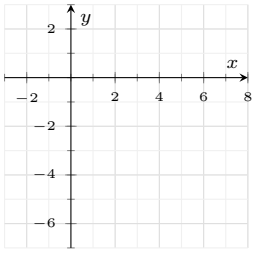
6. At the same moment, one ball is thrown upward from the ground and another from an 18 m balcony. Their heights in metres after t seconds are $h = -5t^2 + 24t$ and $h = -5t^2 + 12t + 18$. When are the balls at the same height, and what is that height?

7. Graph each inequality.

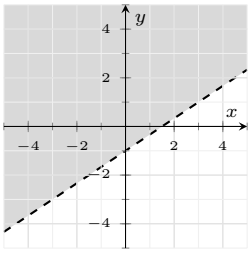
(a) $y \leq -2x + 3$



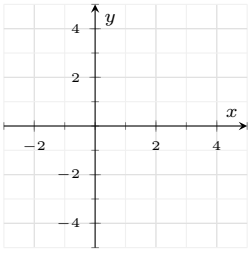
(b) $2x - 3y > 12$



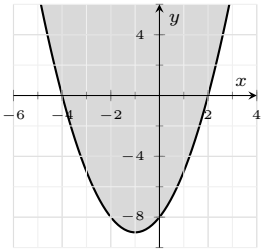
8. The graph shows the solution region of an inequality. The dashed boundary passes through $(0, -1)$ and $(3, 1)$. Write the inequality.



9. Graph $y < -x^2 + 2x + 3$.



10. The boundary of the shaded region is a parabola with x-intercepts -4 and 2 and y-intercept -8. Write the inequality.



11. A grad committee sells muffins for \$2 and coffees for \$3. It needs to raise at least \$150. Let m be the number of muffins and c the number of coffees sold.

(a) Write an inequality, with restrictions, for the situation.

(b) Is selling 30 muffins and 25 coffees enough? Is 45 muffins and 20 coffees?

(c) Explain why the boundary is solid and why only some points in the shaded region make sense.

12. Solve. Write each solution in set notation.

(a) $x^2 - 5x - 14 < 0$ (b) $3x^2 \geq 2 - 5x$ (c) $16 - x^2 > 0$

(d) $x^2 + 4x + 5 \leq 0$ (e) $x^2 - 12x + 36 > 0$

13. Solve $-2x^2 + 3x + 20 \leq 0$ using a sign chart.

14. A ball is thrown upward from a 10 m platform. Its height after t seconds is $h = -5t^2 + 30t + 10$ metres.

(a) When is the ball at least 35 m high, and for how long?

(b) When is the ball below the platform? Round to the nearest hundredth.

15. (a) Explain why $(0, 0)$ cannot be used as a test point for $y < x^2 + 3x$, and choose a better one.

(b) Explain how the solution of $x^2 + 3x < 0$ differs from the solution of $y < x^2 + 3x$, and solve $x^2 + 3x < 0$.

16. The number of solutions of the system $y = x^2 - 6x + 10$ and $y = 2x - 6$ is
A. 0 **B.** 1 **C.** 2 **D.** infinitely many

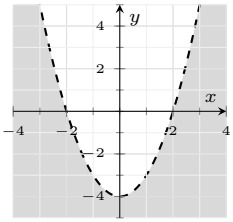
17. When the system $y = 2x^2 - 3x + 1$ and $x + y = 4$ is solved by substitution, the quadratic equation that results is
A. $2x^2 - 4x - 3 = 0$ **B.** $2x^2 - 2x - 3 = 0$ **C.** $2x^2 - 2x + 5 = 0$ **D.** $2x^2 - 4x + 5 = 0$

18. Numerical Response To the nearest hundredth, the positive x-coordinate of a solution of the system $y = x^2 + x - 4$ and $y = 2x + 1$ is _____.

19. The point that is **not** in the solution region of $2x - 5y \geq 10$ is
A. $(5, 0)$ **B.** $(1, -3)$ **C.** $(-1, -2)$ **D.** $(0, -2)$

20. The boundary of the shaded region is the parabola $y = x^2 - 4$. The region is the solution of

A. $y < x^2 - 4$ **B.** $y > x^2 - 4$
C. $y \leq x^2 - 4$ **D.** $y \geq x^2 - 4$



21. The solution of $x^2 + 2x - 35 \geq 0$ is
A. $-7 \leq x \leq 5$ **B.** $x \leq -7$ or $x \geq 5$ **C.** $-5 \leq x \leq 7$ **D.** $x \leq -5$ or $x \geq 7$

22. Numerical Response The number of integers in the solution of $2x^2 - 7x - 15 \leq 0$ is _____.

23. Ravi solves $-3x^2 + 3x + 18 > 0$ with a sign chart using the factors -3, $x + 2$ and $x - 3$. From left to right, the product row of his chart is
A. $+, 0, -, 0, +$ **B.** $-, 0, +, 0, -$ **C.** $-, 0, -, 0, +$ **D.** $+, 0, +, 0, -$

24. Which point is in the solution regions of **both** $y > x^2 - 4$ and $y < x + 2$?
A. $(0, 3)$ **B.** $(3, 4)$ **C.** $(0, 0)$ **D.** $(-3, 0)$

25. Numerical Response A craft business makes and sells n items a week. Its weekly profit, in dollars, is $P = -0.5n^2 + 40n - 600$. The least number of items it must sell so that it does not lose money ($P \geq 0$) is _____.

26. The line $y = kx - 1$ is tangent to the parabola $y = x^2 + 3$. A possible value of k is

A. 2 **B.** 4 **C.** 8 **D.** 16

27. Numerical Response The parabolas $y = x^2 - 2x - 3$ and $y = -x^2 + 4x + 5$ intersect at (a, b) and (c, d). The value of $a + b + c + d$ is _____.
